

Long COVID whole-blood and PBMC programs after preregistration: a cross-assay oxidative-phosphorylation signal, two transparent negatives on ME/CFS separation, and a full gate audit

MEGA-PROGRAM-27, Long COVID lane (builder branch builder-25-ic)

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Abstract

Long COVID (post-acute sequelae of SARS-CoV-2 infection, PASC) still lacks any molecular test; its closest clinical neighbour, myalgic encephalomyelitis/chronic fatigue syndrome (ME/CFS), shares both symptoms and putative biology, and the two are routinely conflated in cohort design. This report runs a disciplined, fully preregistered program on four public cohorts and keeps every negative on the record. First, three fixed-list tests derived from published six-gene panels fail their strict rules: the GSE226260-derived list shows 5/6 expected signs in its own source cohort ($p = .0081$, no Bonferroni pass), fails outright on a NanoString PBMC cohort, and a naive-CD4 cohort cannot measure CXCL8 at all. Second, the preregistered P51 arc trains nine Hallmark-derived modules on GSE226260 whole blood (17 PASC vs 20 recovered) and selects exactly one: oxidative phosphorylation, down-regulated in PASC (permutation $p = .0432$, leave-one-patient-out stable 37/37). Frozen evaluation on the untouched GSE275334 PBMC NanoString cohort shows the oxphos-down signature separates Long COVID from healthy controls cross-assay (E1: $p = .0038$, Hedges $g = 0.958$) but does not separate Long COVID from ME/CFS (E2: $p = .485$, $g = 0.259$); the published six-gene comparator also fails the locked $|g| > 0.8$ bar on that contrast ($g = 0.776$). The locked win rule is not met and is recorded as such. Third, the preregistered P52 arc - designed after an advisory design review and correcting the first arc's anti-conservative one-sided selection to a two-sided, platform-stratified rule - attempts to train an ME/CFS-contrast signature on post-infectious ME/CFS PBMC (GSE251872; 12 cases, 15 controls): no module passes (best two-sided $p = .194$), and the arc ends at the training stage, by preregistration, without touching the evaluation cohort. Fourth, we build the lane's gate evidence honestly: 178 accession-backed record units (record units, not patients or studies), 44 genuinely used external scientific services (42 under the conservative recount), and this manuscript toward the 50-substantive-page gate, whose status is stated exactly wherever it stands. No clinical test, named discovery, or benchmark break is claimed.

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1 Scope, verdict and reproducibility

Scope. This manuscript is the disease-lane report for the Long COVID project of MEGA-PROGRAM-27. It covers: (i) provenance and verification of every dataset the lane has used or credited; (ii) the pre-arc fixed-list tests that motivated preregistration; (iii) the two preregistered analysis arcs, P51 and P52, with their frozen protocols, single executions, and untuned outcomes; (iv) the external-service annotation landscape that contextualises the preregistered gene sets; and (v) the lane’s gate audit against the program’s 120-record, 40-service and 50-substantive-page floors.

Verdict, stated up front. The lane holds one preregistered positive and two preregistered negatives. The positive: an oxidative-phosphorylation down-regulation signature, selected under a frozen protocol in whole blood, transports across assay platform and cell mixture to separate Long COVID from healthy controls in PBMC (E1, $p = .0038$). The negatives: that signature does not distinguish Long COVID from ME/CFS (E2, $p = .485$), and a second, better-powered-design arc aimed directly at the ME/CFS contrast found no trainable module signal at all (best two-sided $p = .194$). We claim the positive exactly as far as the preregistration supports: a reproducible chronic-state signal against healthy controls, not a disease-specific discriminator and not a diagnostic. The ME/CFS separation question - the lane’s declared differentiator in its novelty plan - is, after two disciplined attempts, unresolved: the first arc’s signatures fail it, and the second arc shows the failure is not fixed by switching training cohorts at this sample size.

Reproducibility. Every dataset is stored in-repo under `data/geo/` with SHA-256 locks asserted at the top of every analysis script. Every gated test has its preregistration committed and pushed before its first execution (`projects/long_covid/prereg/`). Every analysis script runs end-to-end from the locked data with a fixed seed and writes a JSON result committed alongside. Advisory consultations (judge mode) are preserved verbatim with conversation URLs in `redirect/`. One implementation bug occurred (gzip handling in the P52 trainer); it failed before producing any result, is disclosed in the redirect log, and the fixed script’s output is the single training execution the preregistration allows.

2 Disease background and the ME/CFS adjacency problem

Long COVID denotes persistent symptoms following SARS-CoV-2 infection that cannot be explained by an alternative diagnosis; fatigue, post-exertional malaise, cognitive dysfunction and autonomic features dominate. Its case definition is clinical and exclusionary, which makes a molecular discriminator valuable in principle and dangerous in practice: any blood signal that merely tracks chronic illness will appear to “work” against healthy controls while adding nothing over clinical assessment. The strongest version of the problem is ME/CFS. ME/CFS is likewise a post-infectious, exclusion-diagnosed syndrome with overlapping symptomatology; a nontrivial fraction of Long COVID patients meet ME/CFS criteria. A biomarker claim for Long COVID that has never been tested against ME/CFS controls is, at best, a claim about chronic unwellness.

This is why the lane’s novelty plan names the ME/CFS contrast as the actual contribution to attempt, and why both preregistered arcs are built around it: P51 asks whether a discovered signature separates Long COVID from ME/CFS, not merely from health; P52 asks whether a signature trained on ME/CFS itself separates the two. The honest scientific prior, which the advisory review of the P52 design stated plainly, is that these conditions may be molecularly continuous in blood; the program’s job is to test that falsifiably, not to assume separability.

3 Data acquisition and provenance

3.1 Datasets and roles

Four GEO series are locked in-repo. **GSE226260** (whole-blood RNA-seq; source study Nature Immunology 2025) contributes the P51 training cohort: 17 PASC vs 20 recovered donors, one sample per person

selected under the lane’s sample-map rules from 103 libraries (`results/longcovid_p36_samples.csv`); the same series is the source of the published six-gene panel (JAK1, CXCL8, BCL2L1, OSM, MAP3K8, STAT3) used throughout as the named comparator. **GSE275334** (PBMC, 635-gene NanoString panel) contributes the locked evaluation cohort: 23 Long COVID, 9 ME/CFS, 15 healthy controls (`results/longcovid_p37_samples.csv`) - to our knowledge the only public cohort carrying Long COVID and ME/CFS arms on one platform, which is why it is reserved for evaluation and never trained on. **GSE334857** (naive CD4 T-cells) contributes the p38 coverage test. **GSE251872** (Walitt deep-phenotyping post-infectious ME/CFS, baseline PBMC RNA-seq) contributes the P52 training cohort: 12 cases vs 15 healthy volunteers across two sequencing platforms (3/7 and 9/8 case/control splits), a confound the P52 preregistration handles by within-platform z-scoring and platform-stratified permutation.

3.2 Verification

All files carry SHA-256 digests asserted in-script before any processing; sample-to-group assignments are taken from series-matrix titles and characteristics, not from secondary citations. The GSE251872 group split (12/15) and platform split were re-derived from the two platform-specific series matrices and are re-asserted in the training script, which fails closed if the composition ever changes.

4 Pre-arc fixed-list tests: three negatives that forced preregistration

Before any preregistered arc, the lane tested the obvious cheap hypotheses - published fixed gene lists - under strict all-gene rules (`results/longcovid_p36_result.json`, p37, p38). The GSE226260 six-gene list, evaluated in its own source cohort, shows 5/6 expected signs ($p = .0081$) but no gene passes a Bonferroni bar: a weak, post-selected signal exactly where it was discovered. The same list fails its strict rule outright on GSE275334. The naive-CD4 cohort cannot measure CXCL8 (the list’s anchor gene) above detection, so the third test is uninterpretable by construction and recorded as a coverage failure, not a result. These negatives are why the lane moved to frozen-protocol module arcs: post-selected six-gene lists do not survive contact with independent cohorts, and any discovered signature must be selected where it is trained and judged where it has never been seen.

5 Preregistered arc P51: the oxidative-phosphorylation signal

5.1 Frozen design

The P51 preregistration (committed at 42deadfa before any run) fixed: nine Hallmark-derived candidate modules L1–L9 (`prereg/lc_p51_module_genesets.json`); training on GSE226260 only; module eligibility bars (≥ 20 genes measurable in training, ≥ 10 on the 635-gene evaluation panel, locked a priori because the panel is small); selection by Cohen’s D plus permutation testing with leave-one-patient-out sign stability; and a frozen evaluation on GSE275334 comprising E1 (signature vs healthy, one-sided in the trained direction, pass $p < .025$), E2 (signature vs ME/CFS, two-sided $p < .05$ and $|g| > 0.8$), the same two tests for the six-gene comparator on the same subjects with the same machinery, and a win rule requiring E1 and E2 to pass and the signature’s E2 effect to exceed the comparator’s.

5.2 Training result

One module of nine was selected: L7, oxidative phosphorylation, down-regulated in PASC vs recovered ($D = -0.356$, permutation $p = .0432$, sign stable in all 37 leave-one-out folds; `results/lc_p51_train_result.json`). The other eight modules show permutation p between .10 and .29 - no second signal. Twelve of the module’s genes are on the evaluation panel, clearing the locked ≥ 10 bar.

5.3 Evaluation result and the honest ledger

On GSE275334 (results/lc_p51_eval_result.json): **E1 passes** - the oxphos-down signature separates Long COVID from healthy controls cross-assay ($D = 0.343$, $p = .0038$, $g = 0.958$). **E2 fails** - it does not separate Long COVID from ME/CFS ($D = 0.113$, $p = .485$, $g = 0.259$). The six-gene comparator is stronger on both tests ($p = .0018$, $g = 1.051$ against healthy; $p = .0473$, $g = 0.776$ against ME/CFS) yet itself fails the locked $|g| > 0.8$ bar on the ME/CFS contrast. The win rule is not met; the arc is recorded as a ledger outcome, untuned.

5.4 The anti-conservatism note, carried forward

The P51 training permutation was implemented one-sided in each module's observed direction; because direction was not fixed a priori, this is anti-conservative relative to a truly prespecified test. The flaw is disclosed in the result commit, the training result was not re-run (no tuning into a different answer), and the correction was mandated into the P52 preregistration: two-sided selection, stated below.